

Regresi: Estimasi Redshift Fotometri

SK 5004 MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

MEET OUR MEMBERS



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DAFTAR ISI

01 Problem Definition

02 Proposed Solution

03 Result

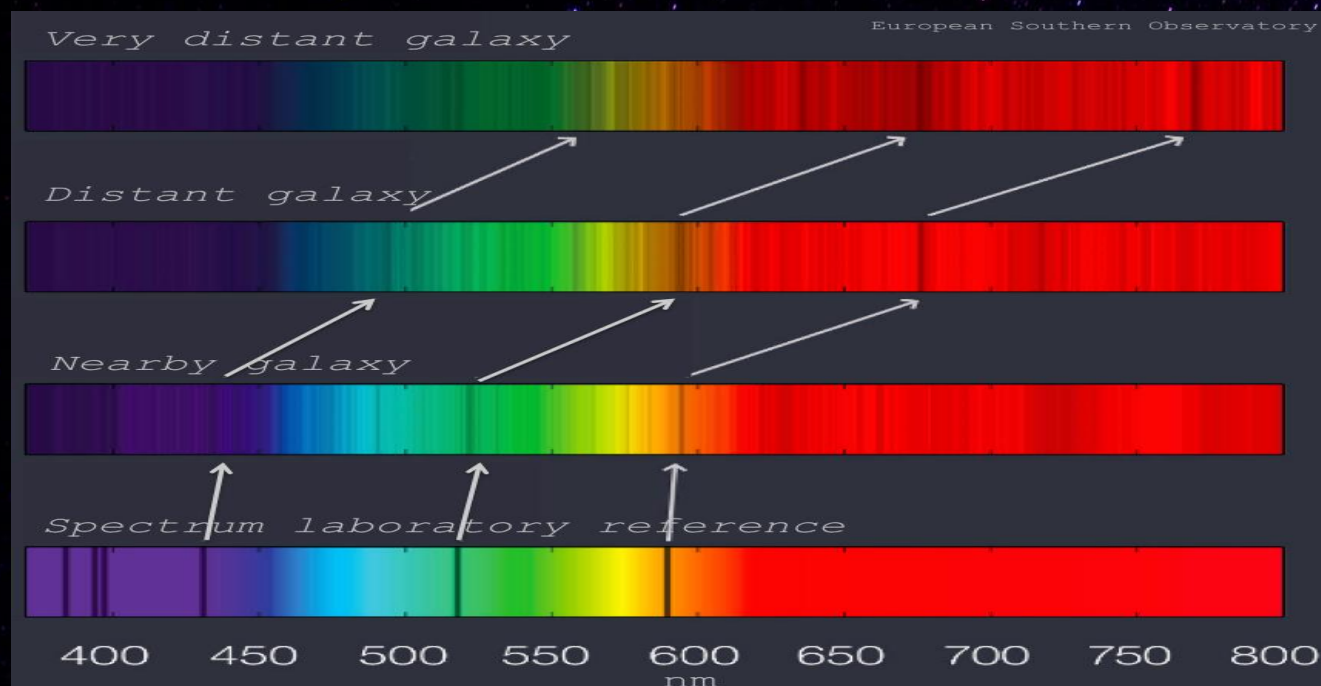
04 Discussion



Redshift

Redshift adalah hasil dari efek Doppler. Efek Doppler adalah pergeseran panjang gelombang atau frekuensi dari gelombang yang dipancarkan suatu sumber yang bergerak relatif terhadap pengamat.

- Redshift -> pergeseran menuju panjang gelombang yang lebih panjang
- Blueshift -> pergeseran menuju panjang gelombang yang lebih pendek



Problem

Redshift biasanya diukur menggunakan Spektroskopi, namun terdapat batasan diantaranya :

1. Butuh waktu lama untuk mendapatkan spektrum objek
2. Jumlah observasi terbatas
3. Instrumen mahal



Solution

Pendekatan menggunakan machine learning dari data fotometri untuk memprediksi redshift sudah dilakukan oleh beberapa penelitian pada Tabel disamping, Karena pendekatan dengan ML tidak memerlukan pengetahuan fisis, kemudian lebih cepat dan scalable.

Oleh karena itu, kita mencoba melakukan optimasi untuk menurunkan nilai outlier (η) dengan menggunakan model yang lebih *robust* terhadap outliers.

Sumber	Model	Mean Bias (Δz)	σ	η (3σ)	η (bias > 0.05)
Beck et al. (2016)	KNN	5×10^{-5}	0.0205	4.10%	-
Zhou et al. (2018)	Linear Local Regression	2.3×10^{-4}	0.019	4.20%	-
M.Treyer et al. (2023)	CNN	1.5×10^{-3}	0.014	-	4%

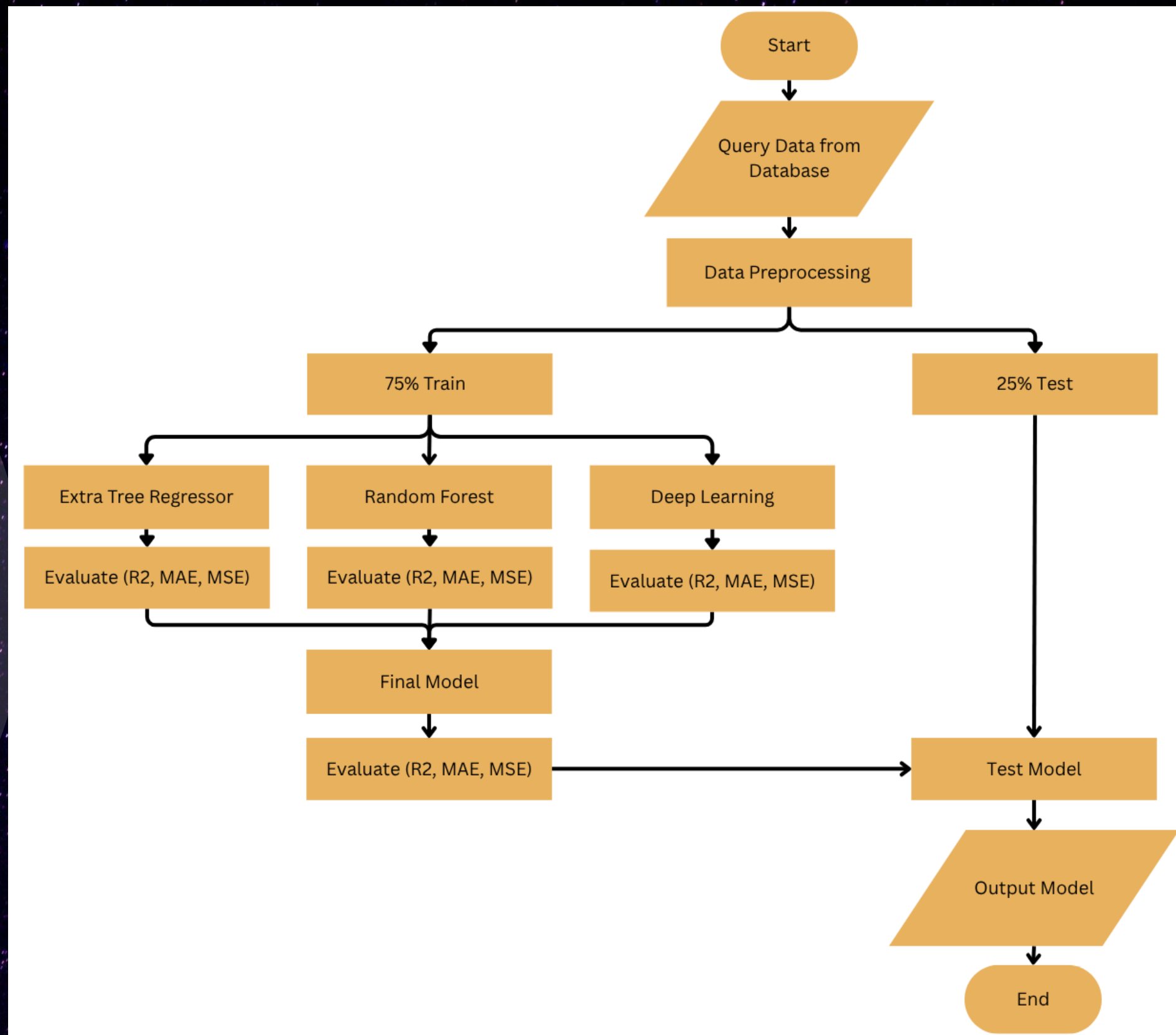


Raw Data

	specObjID	dered_u	dered_g	dered_r	dered_i	dered_z	redshift	zErr	class	zWarning
0	299489676975171584	16.91713	15.31460	14.55245	14.14041	13.83297	0.021275	0.000009	GALAXY	0
1	299489677444933632	16.91713	15.31460	14.55245	14.14041	13.83297	0.021222	0.000009	GALAXY	0
2	299489951853078528	20.18201	18.45391	17.33205	16.89840	16.59294	0.213925	0.000052	GALAXY	0
3	299489952322840576	20.97757	18.85416	17.52073	17.00673	16.65681	0.203783	0.000039	GALAXY	0
4	299490501608892416	20.53679	18.46119	17.18082	16.61607	16.14976	0.126554	0.000062	GALAXY	0
...	...	...	...	...	...	...	...	...	...	...
299995	1012345963295041536	18.82024	17.81306	17.41734	17.19825	17.11883	0.041180	0.000019	GALAXY	0
299996	1012347062806669312	26.08037	20.67783	18.87760	18.23855	17.81052	0.410612	0.000119	GALAXY	0
299997	1012347337684576256	16.79983	16.23388	16.43251	16.54471	16.54948	0.027269	0.000007	GALAXY	16
299998	1012348162318297088	20.34428	19.81071	19.02421	18.62903	18.45377	0.184791	0.000034	QSO	0
299999	1012348437196204032	22.96375	20.86570	19.12657	18.44198	18.07430	0.409397	0.000096	GALAXY	0

Method

Flowchart





Train Data

	redshift	class	u-g	g-r	r-i	i-z
0	0.021275	0	1.60253	0.76215	0.41204	0.30744
1	0.021222	0	1.60253	0.76215	0.41204	0.30744
2	0.213925	0	1.72810	1.12186	0.43365	0.30546
3	0.203783	0	2.12341	1.33343	0.51400	0.34992
4	0.126554	0	2.07560	1.28037	0.56475	0.46631
...	...	...	...	...	...	...
299994	0.050493	0	1.83602	0.78007	0.37262	0.27744
299995	0.041180	0	1.00718	0.39572	0.21909	0.07942
299996	0.410612	0	5.40254	1.80023	0.63905	0.42803
299998	0.184791	1	0.53357	0.78650	0.39518	0.17526
299999	0.409397	0	2.09805	1.73913	0.68459	0.36768

292337 rows × 6 columns

Result Base Model



Modeling

Result

Model	Row Number (Data)	Training time (s)	Prediction time (s)	R ²	MSE	MAE
Deep Convolution Neural Network	237.269	1136.4	3	0.74	1.4×10^{-3}	2×10^{-2}
Random Forrest	237.269	110	2	0.77	1.3×10^{-3}	1.6×10^{-2}
Extra Tree	237.269	30	2.2	0.79	1.1×10^{-3}	1.5×10^{-2}
Ensemble	237.269	1	5	0.8	1.1×10^{-3}	1.6×10^{-2}



Modeling

Hyperparameter Tuning

DIAPLIKASIKAN PADA EXTRA TREES DAN RANDOM FOREST

Genetic Algorithm

Parameters:

Evolusi

- Generation: 20
- Population size: 16
- Individual template range: [1, 100]

Seleksi

- Selection: roulettewheel

Crossover

- Pc (probabilitas crossover): 0.8
- Pe (probabilitas crossover per bit): 0.5
- Crossover: uniformcrossover

Mutasi:

- Pm (probabilitas per individu: 0.1
- Pbm (probabilitas per bit: 0.55
- Alpha: 0.6
- Mutation : Flipbitbigmutation

Fitness Function:

- Fitness: r2_score
- Solution[0]: n_estimators

Extra Trees dan Random Forest:

Best Fitness (0.79), Generation (20), Mean Fitness (0.778)

Random Search

Parameters:

- N Estimators : [100, 300, 500]
- Max Depth : [10, 20, 40, None]
- Min Samples Split: [2, 5, 10]
- Min Samples Leaf: [1, 2, 4],
- Max Features: [sqrt dan log]

Best Parameter:

Extra Trees dan Random Forest:

max_feature (log2), n_estimators: (300), max_depth (default), min_samples_split (default), min_samples_leaf (default)



Result



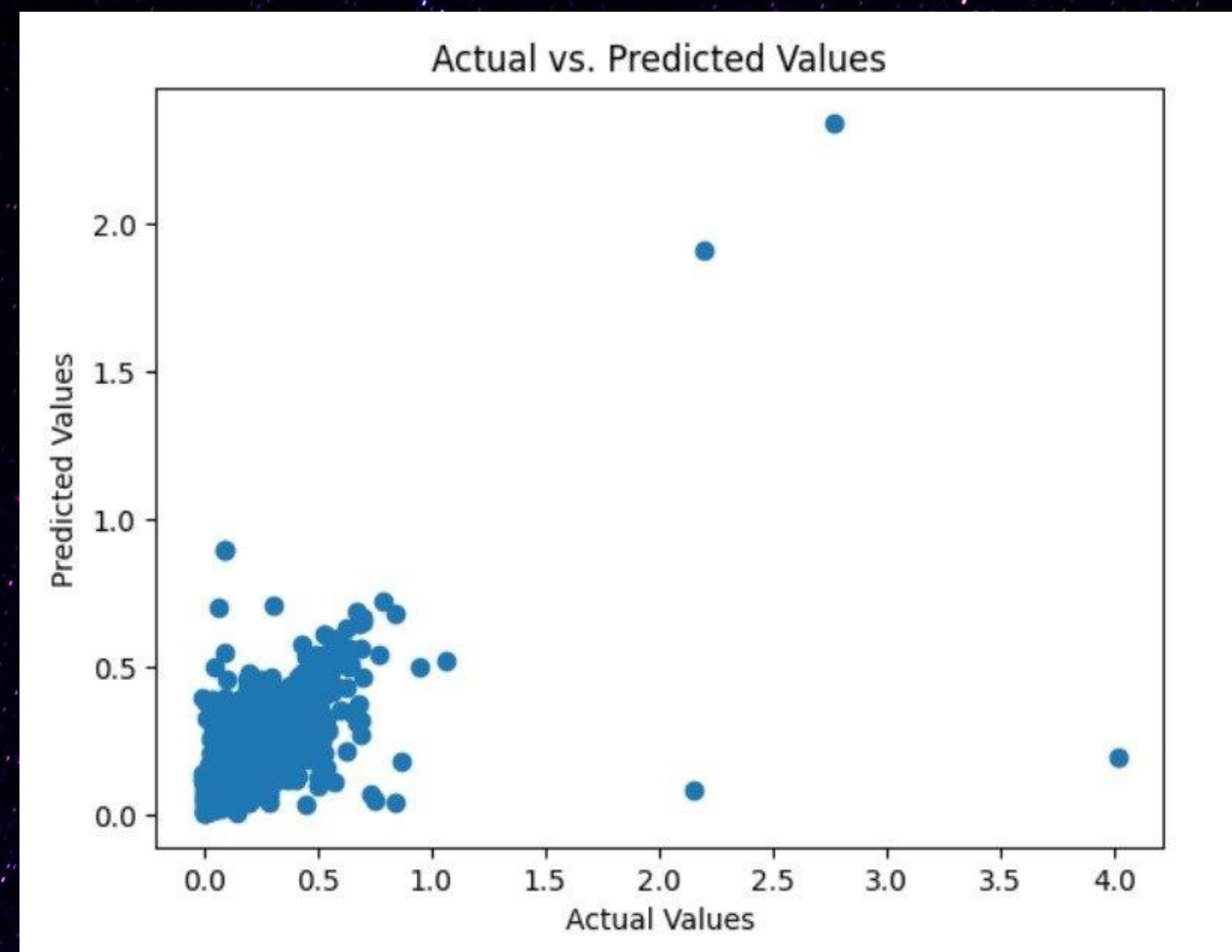
Optimized Result

Model	Row Number (Data)	R ²	MSE	MAE
Deep Convolution Neural Network	237.269	0.74	1.4 x 10 ⁻³	2 x 10 ⁻²
Random Forest	237.269	0.79	1.1 x 10 ⁻³	1.6 x 10 ⁻²
Extra Tree (Random Search)	237.269	0.8	1.1 x 10 ⁻³	1.4 x 10 ⁻²
Extra Tree (GA)	237.269	0.79	1.1 x 10 ⁻³	1.62 x 10 ⁻²
Ensemble	237.269	0.81	1.0 x 10 ⁻³	1.5 x 10 ⁻²



Modeling

Optimized Result





Other Research Comparison

Sumber	Model	Mean Bias (Δz)	σ	η (3σ)	η (median bias > 0.05)
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Zhou et al. (2018)	Linear Local Regression	2.3×10^{-4}	0.019	4.20%	-
M.Treyer et al. (2023)	CNN	1.5×10^{-3}	0.014	-	4%
Our Model	Extra Tree	6.8×10^{-4}	0.023	1.20%	3.30%

$$\Delta z = \frac{z_{pred} - z_{true}}{1 + z_{true}}$$



Timeline

Activities	28 Mei	29 Mei	30 Mei	31 Mei	1 Jun	2 Jun	3 Jun	4 Jun	5 Jun	6 Jun	7 Jun	8 Jun	9 Jun
Hyperparameter Tuning Model													
Output Making													
Publish Output													

$$\Delta z = \frac{z_{pred} - z_{true}}{1 + z_{true}}$$



Work Distribution

1	Data Collection dan Preprocessing (Rizaldi)
2	Data Modeling (Giovaldi & Gia)
3	Hyperparameter Tuning (Gia & Aldi)
4	Presentation (All members)

Output Distribution

1	Youtube (All members)
2	Git Code (Aldiansyah)
3	Medium (Rizaldi dan Gia)
4	OSF (Giovaldi)

$$\Delta z = \frac{z_{pred} - z_{true}}{1 + z_{true}}$$

THANK YOU

